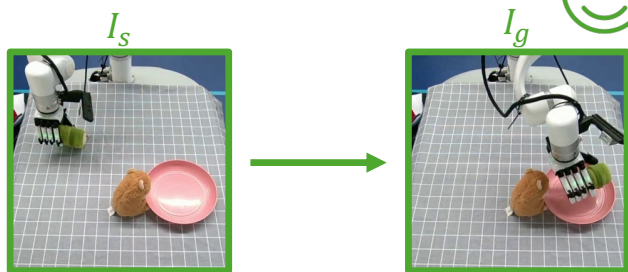


GALA: Geometry-Aware Latent Action Modeling for Vision-Language-Action Model Pretraining across Embodiments

Problem: Image-based LAM?

Case 1: Large movements **perform well** 😊



Case 2: Fine-grained motions? 😞

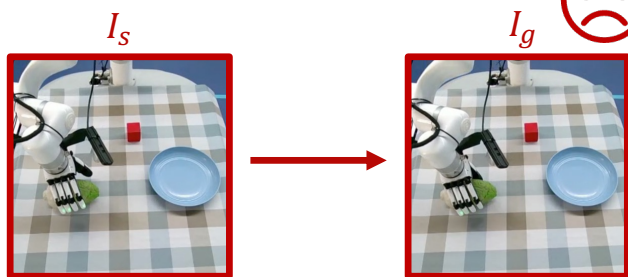


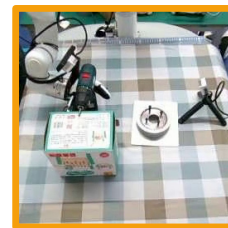
Image-only latent actions are insufficient for modeling **fine-grained** manipulations.

GALA: Augment LAM with Geometry

Visual Transition



I_s



I_g

Point Cloud Geometric Transition



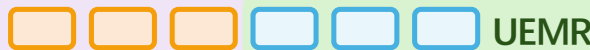
P_s



P_g

GALA

(Geometry-Aware Latent Action)



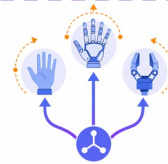
Advantages



Visual & Geometric Transition Modeling

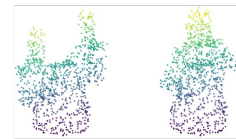


Fine-Grained Motion Capturing

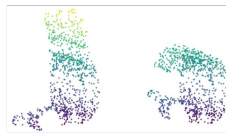


Multi-Embodiment Pretraining

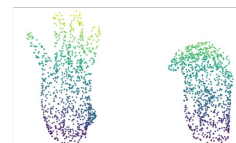
Outcome: Shared Semantics



Gripper



Dexhand 1



Human



Dexhand 2

Geometric Transitions across Embodiments from the Same Latent Action Codes

Outcome: Better Control

Real-World Success Rate (%) ↑

